Tutorial 8: Linear association between quantitative variables

1. In recent years it has been suggested that there is a relationship between global warming and the concentration of carbon dioxide (CO_2). In a study aiming to investigate this relationship, the concentration of CO_2 (in parts per million) and temperatures (in ^oC) for different years were found as follows (data from the Earth Policy Institute):

CO_2:	314	317	320	326	331	339	346	354	361	369
Temperature:	13.9	14.0	13.9	14.1	14.0	14.3	14.1	14.5	14.5	14.4

   (a) Draw a scatter plot of the data and comment on the association between CO_2 concentration and temperature.

   (b) Calculate the correlation coefficient between the two variables.

   (c) Test the hypothesis that there is no linear correlation between CO_2 concentration and temperature.

   (d) Does your analysis here prove that high concentration of CO_2 causes global warming?

2. For a particular insurance company, a sample of ten claims and payments on household contents policies is taken. The table gives the claim and payment (both in £100) for each of the claims considered.

Claim x :	2.80	3.40	3.60	5.00	5.80	6.20	6.80	7.00	7.60	8.60
Payment y :	3.06	2.82	3.78	3.92	6.04	5.20	6.74	6.12	7.46	7.60

   (a) Calculate the sample correlation coefficient for the data and comment briefly.

   (b) Test $H_0 : \rho = 0.8$ v $H_1 : \rho > 0.8$.

3. Consider the set of paired data $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ to which we fit the linear regression model:

$$Y_i \sim \mathrm{N}\left(\alpha + \beta x_i, \sigma^2\right),$$

   where the Y_i are independent. Show that the maximum likelihood estimates of of α and β are given by $\hat{\beta} = \frac{s_{xy}}{s_{xx}}$ and $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$.

4. For the situation of Q3, let $\hat{B}$ denote the maximum likelihood estimator of β (i.e. $\hat{B}$ considered to vary randomly over multiple replicates). Show that $\mathbb{E}(\hat{B}) = \beta$ and $\mathbb{V}\mathrm{ar}(\hat{B}) = \frac{\sigma^2}{s_{xx}}$.

5. Consider again the data in Question 1.

   (a) Fit a simple linear regression model to these data, with temperature being the response (y) variable and CO_2 the explanatory variable (x).

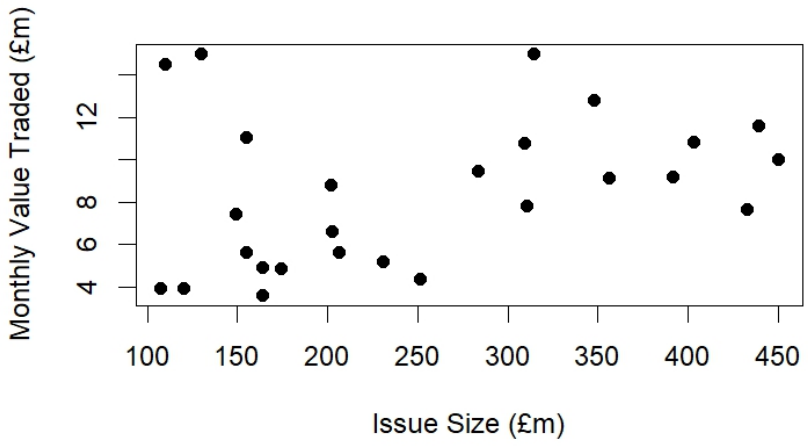
(b) What proportion of the total variation in the data is explained by this model?

(c) Test the hypothesis that the true slope of the regression line is equal to zero at a 1% significance level, i.e. $H_0 : \beta = 0 \;\; v \;\; H_1 : \beta \neq 0$.

(d) The model can be fitted in `R` using `fit.globalw = lm(temp ~ co2)` and the ouput can be seen with the command `summary(fit.globalw)` (where you should put the values of the two variables in `temp` and `co2`). Run the commands in `R` and use the output to verify your answers in (a), (b) and (c).

Exam Style Question

An analyst is instructed to investigate the relationship between the size of a bond issue (x) and its monthly value traded (y). The data for 26 bonds are plotted in the following chart.



(a) Comment on the relationship between issue size and monthly value traded.

[2 marks]

(b) You are given that

$$\sum x_i = 6559.13, \sum x_i^2 = 1967624, \sum y_i = 219.56, \sum y_i^2 = 2166.724, \sum x_i y_i = 58635.97.$$

Determine the correlation coefficient, r between x and y.

[3 marks]

(c) Perform a statistical test to determine if r is significantly different from 0.

[3 marks]